

html, run', imports the required components from the 'reactpy' library. 'html' offers functions to create HTML elements, 'component' is a decorator used to define React components, and 'run' is used to execute the React application.

2. The decorator '@component' designates the ensuing function as a React component.
3. 'def Show():' declares the 'Show' function, which symbolises a React component.
4. The line that reads 'return html.h1("Hello, world!")' specifies the return value for the 'Show' component. It creates an HTML heading element ('h1>') with the text 'Hello, world!' using the 'html.h1' function from the 'html' module.
5. The React application is launched with the command 'run(Show, host="192.168.11.19", port=7000)'. The application is launched using the 'run' function from the 'reactpy' library. The root component of the application is passed as the 'Show' component. The IP address and port number on which the application will be hosted are specified by the 'host' and 'port' arguments. In this instance, it will be accessible via port 7000 and IP address '192.168.11.19'.

The code above creates a straightforward React component called 'Show' that renders an HTML heading element with the text 'Hello, world!'. After that, the React application is launched and hosted at the given IP address and port.

Creating a 'To-do List' application

We can create a 'To-do List' application with the help of ReactPy as follows:

```
from reactpy import component, html, run
@component
def DataList(items, filter_by_priority=None, sort_by_
priority=False):
    if filter_by_priority is not None:
        items = [i for i in items if i["priority"] <= filter_
by_priority]
    if sort_by_priority:
        items = sorted(items, key=lambda i: i["priority"])
    list_item_elements = [html.li({"key": i["id"]},
i["text"]) for i in items]
    return html.ul(list_item_elements)
@component
def TodoList():
    tasks = [
        {"id": 0, "text": "Read a book", "priority": 0},
        {"id": 1, "text": "Meet Geet", "priority": 0},
        {"id": 2, "text": "Hit the gym", "priority": 2},
        {"id": 3, "text": "Buy egg", "priority": 1},
        {"id": 4, "text": "Clean the house", "priority": 2},
        {"id": 5, "text": "organize office", "priority": 2},
```

```
        {"id": 6, "text": "Do some coding", "priority": 1},
        {"id": 7, "text": "Read a book", "priority": 1},
    ]
    return html.section(
        html.h1("My Todo List"),
        DataList(tasks, filter_by_priority=1, sort_by_
priority=True),
    )
run(TodoList, host="192.168.11.19", port=7000)
```

The output of the To-do List application is shown in Figure 5.



Figure 5: Output of the 'To-do List' application on the browser

Let's move on to something more challenging. You can add images, titles, and captions, and combine them into one function by using various HTML elements:

```
from reactpy import component, html, run
@component
def Title(title):
    return html.h1(title)
@component
def Photo():
    return html.img(
        {
            "src": "https://cdn.media.
amplience.net/i/epammarketplace/open_source_license_
header?version=1684831857337",
            "style": {"width": "40%"},
        }
    )
@component
def PhotographerName(caption):
    return html.h4(caption)
@component
def PhotoViewer():
    return html.section(
        Title("Picture of the day"),
        Photo(),
        PhotographerName("Visos Banner")
    )
run(PhotoViewer, host="192.168.11.19", port=7000)
```